FED. SUP CLASS 5305

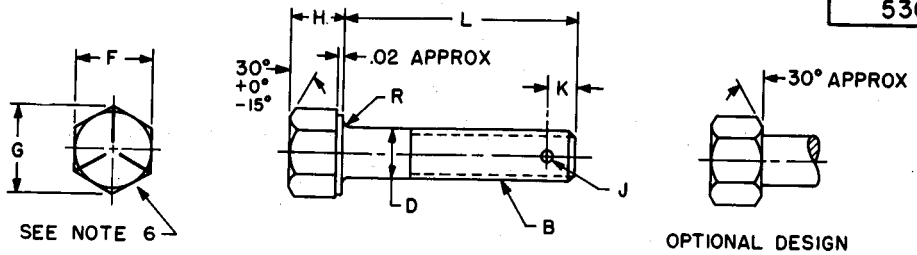



TABLE I

| NOMINAL SIZE | | 1/4 | 5/16 | 3/8 | 7/16 | 1/2 | 5/8 | 3/4 | 7/8 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| B THREAD UNC-2A | | 20 | 18 | 16 | 14 | 13 | 11 | 10 | 9 | 8 |
| D BODY DIAMETER | MAX | .2500 | .3125 | .3750 | .4375 | .5000 | .6250 | .7500 | .8750 | 1.0000 |
| | MIN | .2450 | .3065 | .3690 | .4305 | .4930 | .6170 | .7410 | .8660 | .9900 |
| F WIDTH ACROSS FLATS | MAX | .4375 | .5000 | .5625 | .6250 | .7500 | .9375 | 1.1250 | 1.3125 | 1.5000 |
| | MIN | .4280 | .4890 | .5510 | .6120 | .7360 | .9220 | 1.1000 | 1.2850 | 1.4690 |
| G WIDTH ACROSS CORNERS | MAX | .505 | .577 | .650 | .722 | .866 | 1.083 | 1.299 | 1.516 | 1.732 |
| | MIN | .488 | .557 | .628 | .698 | .840 | 1.051 | 1.254 | 1.465 | 1.675 |
| H HEAD HEIGHT | MAX | .163 | .211 | .243 | .291 | .323 | .403 | .483 | .563 | .627 |
| | MIN | .150 | .195 | .226 | .272 | .302 | .378 | .455 | .531 | .591 |
| J HOLE DIAMETER | +.006 -.002 | .076 | .076 | .107 | .107 | .107 | .140 | .140 | .140 | .140 |
| K ±.010 | | .109 | .125 | .156 | .156 | .172 | .188 | .219 | .250 | .250 |
| R RADIUS | MAX | .025 | .025 | .025 | .025 | .025 | .045 | .045 | .065 | .095 |
| | MIN | .015 | .015 | .015 | .015 | .015 | .020 | .020 | .040 | .060 |
| MIN TENSILE STRENGTH - LBS | | 3,800 | 6,300 | 9,300 | 12,800 | 17,000 | 27,100 | 40,100 | 55,400 | 72,700 |

| L LENGTH | TOLERANCE SIZE 3/4 AND UNDER | TOLERANCE SIZE 7/8 AND OVER | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| .375 .438 .500 .562 .625 .750 .875 1.000 | +0 -.031 | --- | 301 302 303 304 305 306 307 308 | 327 328 329 330 331 332 333 334 | 355 356 357 358 359 360 | 384 385 386 | 410 411 412 | 433 434 435 | 457 | | |
| 1.250 1.500 1.750 2.000 | +0 -.062 | +0 -.125 | 309 310 311 312 | 335 336 337 338 | 361 362 363 364 | 387 388 389 390 | 413 414 415 416 | 436 437 438 439 | 458 459 460 461 | 482 483 484 485 | 506 507 508 509 |
| 2.250 2.500 2.750 3.000 3.250 3.500 3.750 4.000 4.250 4.500 4.750 5.000 5.500 6.000 | +0 -.094 | +0 -.188 | 313 314 315 316 317 318 319 320 321 322 323 324 | 339 340 341 342 343 344 345 346 347 348 349 350 | 365 366 367 368 369 370 371 372 373 374 375 376 | 391 392 393 394 395 396 397 398 399 400 401 402 | 417 418 419 420 421 422 423 424 425 426 427 428 429 430 | 440 441 442 443 444 445 446 447 448 449 450 451 452 453 | 462 463 464 465 466 467 468 469 470 471 472 473 474 475 | 486 487 488 489 490 491 492 493 494 495 496 497 498 499 | 510 511 512 513 514 515 516 517 518 519 520 521 522 523 |

NOTE: For notes and Interchangeability Table see sheet 2.

(D) For changes see sheet 2.

REVIEWER: AV, IS, AT, NSA, OS, 85
USER: ME, GL, MC, MI, YD, AS

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 16 JUL 58 REVISED (A) 16 FEB 65 (B) 16 JUL 65 (C) 14 OCT 68 (D) 1 MAY 1969

| P.A. WC | TITLE | MILITARY STANDARD |
|---|---|---|
| Other Cust SH 82 | SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), SHANK DRILLED FOR COTTER PIN, STEEL, GRADE 5, CADMIUM PLATED, UNC-2A | MS51105 |
| PROCUREMENT SPECIFICATION FF-S-85 | SUPERSEDES: MS51103, BCAX5, BCAX5-1, BUBX20, BUBX20-1 and BUBX21 | SHEET 1 OF 2 |

DD FORM 1 SEP 57 672-1 (Coordinated)

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NOTES:

1. MATERIAL: Medium carbon steel, grade 5, in accordance with procurement specification. Minimum ultimate tensile strength: 120,000 psi.
2. PROTECTIVE COATING: Cadmium plating in accordance with procurement specification.
3. THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H28.
4. THREAD LENGTH: Minimum thread length shall be twice the diameter plus 1/4 inch. The tolerance shall be plus 3/16 inch or 2-1/2 threads, whichever is greater. On screws that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads.
5. DIMENSIONS: All dimensions are in inches, unless otherwise specified.
6. (D) HEAD MARKING: Heads shall be marked in accordance with MIL-HDBK-131. Raised markings are preferred, stampings are acceptable.
7. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51105-301.
8. Referenced documents shall be of the issue in effect on date of invitations for bid.
9. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

## INTERCHANGEABILITY TABLE

The following table shows the interchangeability relationship between the former dash numbers of MS51103, MS51105, and the present dash numbers. Existing stock of screws may be used until depleted. Use only the screws shown as present dash numbers in new design.

TABLE II

| DASH NUMBERS | | | DASH NUMBERS | | | DASH NUMBERS | | | DASH NUMBERS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FORMER | | PRESENT | FORMER | | PRESENT | FORMER | | PRESENT | FORMER | | PRESENT |
| MS51103 | MS51105 | MS51105 | MS51103 | MS51105 | MS51105 | MS51103 | MS51105 | MS51105 | MS51103 | MS51105 | MS51105 |
| 1 | 1 | 301 | 50 | 50 | 350 | 115 | 115 | 415 | 167 | 167 | 467 |
| 2 | 2 | 302 | 55 | 55 | 355 | 116 | 116 | 416 | 168 | 168 | 468 |
| 3 | 3 | 303 | 56 | 56 | 356 | 117 | 117 | 417 | 169 | 169 | 469 |
| 4 | 4 | 304 | 57 | 57 | 357 | 118 | 118 | 418 | 170 | 170 | 470 |
| 5 | 5 | 305 | 58 | 58 | 358 | 119 | 119 | 419 | 171 | 171 | 471 |
| 6 | 6 | 306 | 59 | 59 | 359 | 120 | 120 | 420 | 172 | 172 | 472 |
| 7 | 7 | 307 | 60 | 60 | 360 | 121 | 121 | 421 | 173 | 173 | 473 |
| 8 | 8 | 308 | 61 | 61 | 361 | 122 | 122 | 422 | 174 | 174 | 474 |
| 9 | 9 | 309 | 62 | 62 | 362 | 123 | 123 | 423 | 175 | 175 | 475 |
| 10 | 10 | 310 | 63 | 63 | 363 | 124 | 124 | 424 | 182 | 182 | 482 |
| 11 | 11 | 311 | 64 | 64 | 364 | 125 | 125 | 425 | 183 | 183 | 483 |
| 12 | 12 | 312 | 65 | 65 | 365 | 126 | 126 | 426 | 184 | 184 | 484 |
| 13 | 13 | 313 | 66 | 66 | 366 | 127 | 127 | 427 | 185 | 185 | 485 |
| 14 | 14 | 314 | 67 | 67 | 367 | 128 | 128 | 428 | 186 | 186 | 486 |
| 15 | 15 | 315 | 68 | 68 | 368 | 129 | 129 | 429 | 187 | 187 | 487 |
| 16 | 16 | 316 | 69 | 69 | 369 | 130 | 130 | 430 | 188 | 188 | 488 |
| 17 | 17 | 317 | 70 | 70 | 370 | 133 | 133 | 433 | 189 | 189 | 489 |
| 18 | 18 | 318 | 71 | 71 | 371 | 134 | 134 | 434 | 190 | 190 | 490 |
| 19 | 19 | 319 | 72 | 72 | 372 | 135 | 135 | 435 | 191 | 191 | 491 |
| 20 | 20 | 320 | 73 | 73 | 373 | 136 | 136 | 436 | 192 | 192 | 492 |
| 21 | 21 | 321 | 74 | 74 | 374 | 137 | 137 | 437 | 193 | 193 | 493 |
| 22 | 22 | 322 | 75 | 75 | 375 | 138 | 138 | 438 | 194 | 194 | 494 |
| 23 | 23 | 323 | 76 | 76 | 376 | 139 | 139 | 439 | 195 | 195 | 495 |
| 24 | 24 | 324 | 84 | 84 | 384 | 140 | 140 | 440 | 196 | 196 | 496 |
| 27 | 27 | 327 | 85 | 85 | 385 | 141 | 141 | 441 | 197 | 197 | 497 |
| 28 | 28 | 328 | 86 | 86 | 386 | 142 | 142 | 442 | 198 | 198 | 498 |
| 29 | 29 | 329 | 87 | 87 | 387 | 143 | 143 | 443 | 199 | 199 | 499 |
| 30 | 30 | 330 | 88 | 88 | 388 | 144 | 144 | 444 | 206 | 206 | 506 |
| 31 | 31 | 331 | 89 | 89 | 389 | 145 | 145 | 445 | 207 | 207 | 507 |
| 32 | 32 | 332 | 90 | 90 | 390 | 146 | 146 | 446 | 208 | 208 | 508 |
| 33 | 33 | 333 | 91 | 91 | 391 | 147 | 147 | 447 | 209 | 209 | 509 |
| 34 | 34 | 334 | 92 | 92 | 392 | 148 | 148 | 448 | 210 | 210 | 510 |
| 35 | 35 | 335 | 93 | 93 | 393 | 149 | 149 | 449 | 211 | 211 | 511 |
| 36 | 36 | 336 | 94 | 94 | 394 | 150 | 150 | 450 | 212 | 212 | 512 |
| 37 | 37 | 337 | 95 | 95 | 395 | 151 | 151 | 451 | 213 | 213 | 513 |
| 38 | 38 | 338 | 96 | 96 | 396 | 152 | 152 | 452 | 214 | 214 | 514 |
| 39 | 39 | 339 | 97 | 97 | 397 | 153 | 153 | 453 | 215 | 215 | 515 |
| 40 | 40 | 340 | 98 | 98 | 398 | 157 | 157 | 457 | 216 | 216 | 516 |
| 41 | 41 | 341 | 99 | 99 | 399 | 158 | 158 | 458 | 217 | 217 | 517 |
| 42 | 42 | 342 | 100 | 100 | 400 | 159 | 159 | 459 | 218 | 218 | 518 |
| 43 | 43 | 343 | 101 | 101 | 401 | 160 | 160 | 460 | 219 | 219 | 519 |
| 44 | 44 | 344 | 102 | 102 | 402 | 161 | 161 | 461 | 220 | 220 | 520 |
| 45 | 45 | 345 | 110 | 110 | 410 | 162 | 162 | 462 | 221 | 221 | 521 |
| 46 | 46 | 346 | 111 | 111 | 411 | 163 | 163 | 463 | 222 | 222 | 522 |
| 47 | 47 | 347 | 112 | 112 | 412 | 164 | 164 | 464 | 223 | 223 | 523 |
| 48 | 48 | 348 | 113 | 113 | 413 | 165 | 165 | 465 | | | |
| 49 | 49 | 349 | 114 | 114 | 414 | 166 | 166 | 466 | | | |

REVIEWER:AV, IS, AT, NSA, OS, 85
USER:ME, GL, MC, MI, YD,AS

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APPROVED 16 JUL 58 REVISED (B) 16 JUL 65 (C) 14 OCT 68 (D) FOR CHANGES SEE SHEET 2.

P.A. WC
Other Cust SH 82

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), SHANK DRILLED FOR COTTER PIN, STEEL, GRADE 5, CADMIUM PLATED, UNC-2A

MILITARY STANDARD
MS 51105
SHEET 2 OF

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDES: MS51103, BCAX5, BCAX5.1, BUBX20, BUBX20.1 and BUBX21

DD FORM 1 SEP 67 672-1 (Coordinated)
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